JOHOR MATRICULATION ENGINEERING COLLEGE

MATHLETICS SEM 1

SESI 2022/2023

MATHEMATICS 1

INSTRUCTIONS TO STUDENTS

1. *There are TWO sections in the question paper. Section A and Section B.*
2. ***Section A** consists of **3** questions. **Section B** consists of **7** questions.*
3. *Answer all questions.*

LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

Trigonometry
Trigonometri

$$\sin\ (A \pm B) = \sin A \cos B \ \pm\ \cos A \sin B$$

$$\cos\ (A \pm B) = \cos A \cos B \ \mp\ \sin A \sin B$$

$$\tan\ (A \pm B) = \frac{\tan A \ \pm\ \tan B}{1 \ \mp\ \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin\ 2A = 2 \sin A \cos A$$

$$\begin{aligned} \cos 2A &= \cos^2 A - \sin^2 A \\ &= 2\cos^2 A - 1 \\ &= 1 - 2\sin^2 A \end{aligned}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

Quadratic equation $ax^2+bx+c=0$:
Persamaan kuadratik $ax^2+bx+c=0$:

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

Differentiation
Pembezaan

$f(x)$	$f'(x)$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x\tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x\cot x$

$$\frac{dy}{dx}=\frac{dy}{dt}\times\frac{dt}{dx}$$

$$\frac{d^2y}{dx^2}=\frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

Sphere ***Sfera***	$V=\frac{4}{3}\pi r^3$	$S=4\pi r^2$
Right circular cone ***Kon membulat tegak***	$V=\frac{1}{3}\pi r^2h$	$S=\pi r^2+\pi rl$
Right circular cylinder ***Silinder membulat tegak***	$V=\pi r^2h$	$S=2\pi r^2+2\pi rh$

SECTION A [25 *marks*]

***BAHAGIAN* A** [25 *markah*]

This section consists of **3** questions. Answer **all** questions.

Bahagian ini mengandungi **3** *soalan. Jawab* ***semua*** *soalan.*

1 Evaluate the following limits, if they exist.

(a) $\lim_{x \to 0} \frac{\sqrt{2x+9}-3}{x}$

[4 *marks*]

(b) $\lim_{x \to 1} \frac{3x^2-2x-1}{|x-1|}$

[5 *marks*]

2 Find $\frac{dy}{dx}$ for each of the following

(a) $y = 5^{\sqrt{x}}(e^{-x^2})$ [4 *marks*]

(a) $y = \ln\left(\frac{\sin\sqrt{x}}{\sqrt{x^2-1}}\right)$ [5 *marks*]

3. The function *f* is defined by $f(x) = 2x^3 - 12x^2 + 18x + 5$.

(a) Find the relative extrema of the function *f*.

[4 *marks*]

(b) Show that the function *f* has one inflection point at $(2,9)$.

[3 *marks*]

SECTION B [75 *marks*]

***BAHAGIAN* B** [75 *markah*]

This section consists of **7** questions. Answer **all** questions.

Bahagian ini mengandungi **7** *soalan. Jawab* ***semua*** *soalan.*

1 Determine the value of *x* if $\dfrac{\sqrt{7}-xi}{1-\sqrt{7}i}$ is real number and find this real number.

[5 *marks*]

2 (a) Solve $\log_2 2x = \log_4(x+3)$

[4 *marks*]

(b) Find all values of *x* for $\left|\frac{1}{1-x}\right| \geq 3$.

[7 *marks*]

3 The function *f* and *g* are given by

$$f(x) = \frac{2x+3}{2x-3}\ , x \in \mathbb{R}\ , x \neq \frac{3}{2}$$

$$g(x) = x^2 + 2\ , x \in \mathbb{R}$$

a) State the range of $g(x)$.

[1 *mark*]

b) Find an expression for $fg(x)$ in simplest form.

[3 *marks*]

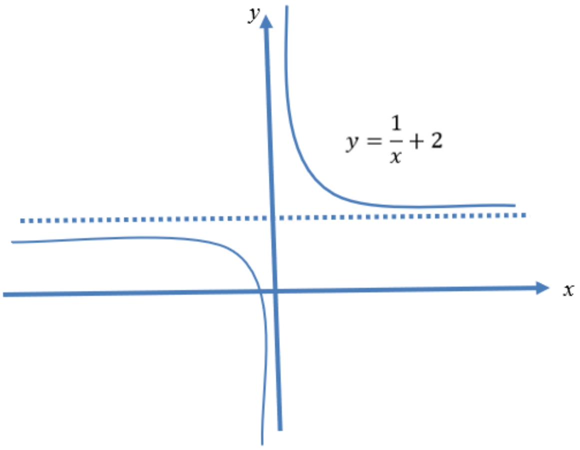
c) Determine an expression for $f^{-1}(x)$ in simplest form.

[3 *marks*]

d) Solve the equation $f(x) = f^{-1}(x)$.

[3 *marks*]

4



The figure above shows the graph of $y = \frac{1}{x} + 2\ ,\ x \neq 0$

a) State the equation of the horizontal asymptote to the curve.

[1 *mark*]

The function *f* is defined

$$f(x) = \frac{1}{x} + 2,\ \ x \in \mathbb{R}\ , x > 1$$

b) State the range of $f(x)$.

[1 *mark*]

c) Obtain an expression for $f^{-1}(x)$.

[3 *marks*]

d) State the domain and range of $f^{-1}(x)$.

[2 *marks*]

e) Sketch the graph of $f(x)$ and $f^{-1}(x)$ on the same axes.

[3 *marks*]

5 Given $f(x) = \dfrac{2x-3}{(x-1)(x+3)}$.

(a) Find the domain of $f(x)$. [1 *mark*]

(b) Find the x-intercept and y-intercept of $f(x)$. [3 *marks*]

(c) Find the vertical asymptote(s) of $f(x)$. [3 *marks*]

(d) Find $\lim_{x\to-\infty} f(x)$ and $\lim_{x\to\infty} f(x)$. Hence, state the horizontal asymptote(s) of $f(x)$.

[4 *marks*]

(e) Hence, sketch the graph of $f(x)$. [3 *marks*]

6

(a) By using the first principle of differentiation, find $\boldsymbol{f'(x)}$ for

$\boldsymbol{f(x)=\sqrt{1-3x}}$**.** [4 *marks*]

(b) Given that $\boldsymbol{x+t=xt}$ and $\boldsymbol{2ty-y^2=3,}$ find $\boldsymbol{\frac{dx}{dt}}$ and $\boldsymbol{\frac{dy}{dt}}$**.** Hence, find the values of $\boldsymbol{\frac{dy}{dx}}$ when $\boldsymbol{x=2.}$ [12 *marks*]

7 Milk is poured into a hemispherical container of radius 8 cm at the rate of $6\pi\ cm^3s^{-1}$. The volume of milk in the container, V cm^3 is given by $V=\pi h^2\left(5-\frac{1}{3}h\right)$, where h is the depth of the milk in cm. At the instant when the depth of the milk is 4 cm, find

(a) The rate of change of the depth of the milk. [4 *marks*]

(b) The rate of change of the radius of the milk's surface area. [5 *marks*]

END OF QUESTION PAPER

KERTAS SOALAN TAMAT